



Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 25 Further Mechanics 1

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Summer 2026

Publication Code 8FMO_25_2606_ER

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General Comments

Overall, learners were able to access all four questions on this paper, and time did not appear to be a limiting factor. Learners generally recalled and used standard formulae accurately and were familiar with the contexts presented across the paper.

Where a printed answer is given, learners should show sufficient detail in their working to warrant the award of all available marks. They should also ensure that their final answer is **exactly the same** as the printed answer.

Learners should show enough working to make their methods clear to the examiner, as stated on the front of the question paper. Correct answers without supporting working may not gain all, or in some cases any, of the available marks.

In questions, the numerical value of g which should be used is 9.8 ms^{-2} . Final answers should then be given to 2 (or 3) significant figures – more accurate answers will be penalised, including fractions but exact multiples of g are usually accepted.

Comments on Individual Questions

Question 1

Part (a)

The first mark on the paper was accessible, as it was awarded for recognising that the rebound speed of the particle after impact with the wall was eu . Substitution of the correct mass into the impulse-momentum principle then earned the method mark.

Stronger responses took account of the change in direction when substituting both velocities into the expression for impulse. Sign errors were a common cause of marks not being awarded when using the impulse-momentum equation. Learners are advised to check the signs in their impulse equations before moving on.

Part (b)

Learners found the change in KE and equated it to the given expression to obtain the correct value for e . Method marks were not gained by those who did not use the correct mass and algebraic processing errors were the main cause of lost accuracy marks.

Question 2

Part (a)

This part of the question tested understanding of the Power equation and N2L for a vehicle moving along a horizontal road. Learners applied both equations successfully to the given context. For some, the biggest challenge in the question was converting between kmh^{-1} and ms^{-1} . This question does not have a calculator warning and those responses who used the relevant calculator function were able to convert between units with ease.

Part (b)

This part increased in challenge as the motion of the vehicle was set on a slope. However, many learners were successfully resolving forces and using N2L to obtain the correct answer. With consistent units being used in this part of the question, some learners were more successful in part (b) than in part (a).

Question 3

Part (a)

More successful learners were able to apply their understanding of the work-energy principle to obtain the required value for U . The formulae for work-done, kinetic energy and gravitational potential energy were familiar to learners. To earn method marks, energy equations must contain all the relevant terms with no additional terms. The most common cause for method marks not being awarded in this part was having an additional term in the energy equation due to double-counting (including both the work-done against weight and the GPE).

Part (b)

This was an 'explain' part of the question and learners demonstrated a good understanding of the relationship between U and W in the question. The written answers provided clear reasoning and generally earned both available marks.

Part (c)

This part increased in challenge as the motion of the particle changed to projectile motion. The question required learners to use energy which is content specific to this module. Those responses who explored projectile motion using only suvat equations were unable to access the marks in this part.

Those who used the conservation of energy principle from B to C usually did so correctly and obtained the correct answer. The most common error arose from solving the correct equation $0.25 V^2 = 16$ to give the incorrect answer, $V = 2$ instead of the correct answer, $V = 8$. Learners are advised to check their answers by substituting back into the first line of their equation.

Question 4

Part (a)

Learners who were well prepared for direct elastic collisions between two particles were generally able to establish equations using conservation of linear momentum and Newton's experimental law. These equations were then processed correctly to obtain the given expression for the speed of P and earn the first five marks out of six available.

The question asked learners to *carefully justify* their reasoning. Some responses included written explanations of how the equations were formed, or how the equations could be solved, but this was not sufficient to justify why the given expression represented the speed of P and gain the sixth mark.

Stronger responses used the condition $e < \frac{1}{2}$, given in the question, to show that the expression obtained was positive. This justified that the expression represented the speed of P .

Part (b)

Learners understood how to obtain the speed of Q from their equations and some had already answered this as part of their solution to (a).

Part (c)

This part of the question was more challenging, as learners were required to consider a subsequent impact with the wall before finding the distance from the wall at which the two particles would collide again.

Responses used a variety of approaches. A common successful approach was to find an expression for the total time taken by P , which travelled at a constant speed throughout. This was then compared with an expression for the total time taken by Q , which travelled a distance d at one speed and a distance x at another speed.

Equating these two times produced an equation that could be rearranged to find the required expression for x in terms of d .